Biostatistics – Section Summary

Study design • Bias • Errors • Diagnostic tests • Effect measures

Critical Care Board Review • Section 36 • MCQ-Oriented

1. Study Design

- Prevalence → **cross-sectional**
- Cohort good for **rare exposures**; case-control for **rare diseases**

2. Bias & Error

- Non-response bias; EGDT single-centre → **selection bias**
- Missing a true effect = **type II error** (underpowered)
- Larger sample ↑ precision (power), not accuracy

3. Measures

- Prevalence = existing cases; incidence = new cases/time
- Longer survival ↑ prevalence
- 95% CI: range likely to contain true value
- Lower prevalence → **PPV ↓, NPV ↑** (Se/Sp unchanged)
- ARDSnet: ARR ~8.8%, **RRR ~22%**, **NNT ≈ 11**

4. Value

- Choosing Wisely: avoid **daily routine blood draws**

5. High-Yield MCQ Pearls

- ☑ Prevalence ↓ → PPV ↓
- ☑ NNT = 1/ARR

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